

Yue Wang

Data Driven Physicist | PhD Candidate at UC Berkeley

Education

University of California, Berkeley

Berkeley, CA

Ph.D. in Physics

May 2027

M.A. in Physics

May 2023

- Awarded the Stuart J. Freedman Memorial Fellowship for top performance and future potential (1 of 1 selected across PhD cohort; \$33,206/semester).
- Selected by faculty as a Berkeley Connect Fellow for excelling mentoring (1 of 3 fellows out of 200+ PhD students in the department; \$30,284/semester).
- Selected by American Physical Society Forum on Graduate Student Affairs (APS FGSA) board members for APS FGSA Travel Award for Excellence in Graduate Research (1 of 5 U.S. recipients; \$500).
- GPA: 3.54/4.0

University of Rochester

Rochester, NY

B.S. in Physics, with highest distinction

May 2021

B.S. in Mathematics, with highest distinction

May 2021

- Stoddard Prize Honorable Mention (top 2 undergraduate theses in physics).
- Continuing Student Scholarship for Undergraduates (competitive merit-based scholarship; \$2,500).
- Elected to Phi Beta Kappa (faculty-nominated; top 10% of graduating class).
- Elected to Sigma Pi Sigma (top students completing ≥ 3 semesters of physics courses).
- Dean's List. Magna Cum Laude. GPA: 3.91/4.0

Selected Research Experience

University of California, Berkeley

Berkeley, CA

Project: Dark Matter Search Data Analysis (LZ Experiment)

September 2021 – July 2025

- Aligned competing priorities from 30+ international institutions to meet accelerated publication timelines.
- Initiated and delivered data-driven presentations to 100+ collaborators using Python data visualization tools.
- Developed and implemented 4 core algorithms powering a world-leading dark matter result published in 2025; these algorithms are being used on large-scale data analysis by 100+ scientists on a daily basis.

University of Rochester

Rochester, NY

Project: Quantitative Modeling of Relativistic Brownian Motion

January 2020 – May 2021

- Authored and defended a 14-page thesis, resulting in **Stoddard Prize** Honorable Mention (top 2 undergraduate theses in physics), showcasing project ownership, technical writing, and communication skills.
- Built a mathematical model for a relativistic stochastic process, implemented a Python simulation (100+ lines of code), derived and tested hypotheses, and delivered statistical characterizations generating novel quantitative insights.

SLAC National Accelerator Laboratory

Menlo Park, CA

Project: Laser Image Data Processing and Analysis

June 2019 – August 2019

- Presented research to SLAC leadership and scientists, resulting in a **1st place in the Intern Shark Tank** and a **2nd place in the poster session**, demonstrating effective verbal and virtual communication.
- Designed and deployed a data driven model to refine 20+ TB of multi-dimensional data, which turned unusable laser images into a *Science* publication.

Leadership Experience

University of California, Berkeley

Berkeley, CA

Lab Safety Coordinator

May 2022 – May 2025

- Led chemical barcoding for 50+ chemicals and streamlined inventory by removing 100% of unused items (38% of total inventory), resulting in a vendor tagging time reduction and worker safety improvements.
- Directed lab safety across 4 spaces, managed 5 researchers to resolve safety hazards, and formalized Standard Operating Procedures (SOPs), achieving **0 violations in city-level audits**.
- Secured external funding source for seismic bracing at \$0 out-of-pocket to preserve research budget while ensuring regulatory compliance, enabling uninterrupted lab operations and a sustainable funding model.

Berkeley Connect Fellow

August 2024 – December 2024

- Selected as 1 of 3 fellows in my department for excelling mentoring (\$30,284 fellowship).
- Coached 29 students via 1:1 and group sessions, securing competitive research placements for students, including a **Department of Energy program** at SLAC National Accelerator Laboratory.
- Launched a standardized group session planning guide adopted by new fellows, driving more efficient session planning and consistent mentoring quality.

Graduate Student Instructor, Department of Physics

January 2022 – May 2024

- Planned and led weekly classes of 45+ students, delivering consistent support and driving engagement.
- Created grading rubrics; coordinated regrade requests with other instructors to ensure consistent evaluation.

University of Rochester

Rochester, NY

Principal Pianist and Soloist, University Orchestras

January 2018 – May 2021

- Aligned musical interpretation with conductor and section leaders and integrated piano section input to deliver 8+ cohesive productions.
- Drove concert promotion, reaching 100+ concert attendees through social media and campus outreach.

Public Talks

- Lightning Talk: “Looking for Dark Matter with Quantum Sensors,” APS Conference for Undergraduate Women and Gender Minorities in Physics. January 26, 2025.
- Guest Speaker: “My Life with Dark Matter,” Lawrence Berkeley National Laboratory Internships Event: Physics at LBNL. September 18, 2024.

Selected Peer Reviewed Publications (out of total of 25)

- J. Aalbers, et al. (LZ Collaboration), “Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment,” *Physical Review Letters*: volume 135, issue 1. July 1, 2025.
- S. Wandel, et al., “Enhanced charge density wave coherence in a light-quenched, high-temperature superconductor,” *Science*: volume 376, issue 6595. May 19, 2022.

Skills and Interests

- The William Lowell Putnam Mathematical Competition, the most prestigious competition in college mathematics: rank 1831.5/3428 (2019), rank 2182.5/4623 (2018)
- Technical Skills: Machine Learning, deep learning, mathematical modeling, linear models, anomaly detection and prediction, time-series data, stochastic process, sampling methods
- Programming Languages: Python, Java, SQL, Unix, C++, LaTeX
- Languages: English, Mandarin
- Interests: Classically trained pianist: performed classical and contemporary works. Awarded **National Music Festival Apprentice Fellowship in 2022 (full tuition and accommodation fellowship)**.